

NETWORK SECURITY LAB REPORT

Suricata IDS Home Lab Attack Detection & Analysis Report

Report Title	Suricata IDS Home Lab — Attack Detection & Analysis
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Classification	UNCLASSIFIED / Educational Use Only
Version	1.0
Environment	VirtualBox — Host-Only Network
IDS Engine	Suricata 8.x

■ **DISCLAIMER:** All attacks and tests documented in this report were performed exclusively in a controlled, isolated virtual lab environment. No real systems, networks, or third-party infrastructure were targeted. This report is intended solely for educational and portfolio purposes.

1. Executive Summary

This report documents the design, deployment, and testing of a Network Intrusion Detection System (IDS) using **Suricata 8.x** within a virtualised home lab. The objective was to author custom detection rules targeting four common attack techniques and verify their effectiveness through controlled simulations performed from a Kali Linux attacker VM.

All four attack scenarios — SYN port scan, ICMP ping sweep, SSH brute force, and reverse shell payload — were **successfully detected**, yielding a 100% detection rate. The findings demonstrate that signature-based IDS rules, when carefully tuned, provide reliable first-line visibility into common network threats.

Key Findings	Value
Attacks Simulated	4
Attacks Detected	4 (100%)
Custom Rules Authored	6 (4 core + 2 bonus)
False Positives Observed	0
Lab Duration	~4 hours

2. Lab Environment

The lab was built entirely within VirtualBox using a Host-Only Network adapter, ensuring complete isolation from any production or external networks. Two virtual machines were configured — one acting as the monitored defender running Suricata, and one acting as the attacker running Kali Linux.

Component	Detail
Hypervisor	VirtualBox 7.x
Defender OS	Ubuntu Server 22.04 LTS
Defender IP	192.168.56.101
Defender NIC	enp0s3 (Host-Only)
Attacker OS	Kali Linux 2024.x
Attacker IP	192.168.56.102
IDS Engine	Suricata 8.0.5
Network Subnet	192.168.56.0/24 (Host-Only)
Log Format	fast.log + eve.json

Network Topology

The diagram below illustrates the logical layout of the lab network. All traffic between attacker and defender traversed the Host-Only interface, monitored in real-time by Suricata.

```
[KALI LINUX – Attacker]
192.168.56.102
|
Host-Only Network 192.168.56.0/24
|
[UBUNTU SERVER – Defender]
192.168.56.101 | Suricata IDS
/var/log/suricata/fast.log
```

3. Tools & Methodology

Tool	Version	Purpose
Suricata	8.0.5	IDS engine — rule matching, alerting, eve.json logging
Nmap	7.94	SYN port scan and ICMP ping sweep simulation
Hydra	9.5	SSH brute-force credential attack simulation
Netcat	1.219	Reverse shell payload transmission simulation
jq	1.6	JSON parsing of eve.json structured logs
Wireshark	4.x	Packet-level traffic capture and analysis

Methodology

- Configure Suricata with HOME_NET set to the Host-Only subnet
- Author custom detection rules targeting specific attack signatures
- Launch each attack from the Kali VM against the Ubuntu defender
- Verify alerts appear in fast.log and eve.json within expected timeframes
- Document findings and produce defensive recommendations

4. Custom Detection Rules

Six custom rules were authored for this lab. Rules 1000001–1000004 target the four primary attack scenarios. Rules 1000005–1000006 are bonus rules covering OS fingerprinting probes and oversized ICMP packets.

SID	Rule Name	Proto	Detection Logic
1000001	SYN Port Scan	TCP	Threshold: 20 SYN pkts / 2s from same source
1000002	ICMP Ping Sweep	ICMP	Threshold: 5 ICMP echo req / 2s from same source
1000003	SSH Brute Force	TCP	Threshold: 5 connections to port 22 / 60s
1000004	Reverse Shell	TCP	Content match: /bin/sh payload in packet

SID	Rule Name	Proto	Detection Logic
1000005	OS Fingerprint Probe	TCP	Bonus: unusual TCP flag combo SF
1000006	Oversized ICMP	ICMP	Bonus: ICMP echo request dsize > 1000 bytes

Sample Rule — SSH Brute Force (SID 1000003)

```
alert tcp any any -> $HOME_NET 22 ( msg:"SSH Brute Force Attempt"; flow:to_server;  
threshold: type both, track by_src, count 5, seconds 60; classtype:attempted-admin;  
sid:1000003; rev:1;)
```

5. Attack Scenarios & Detection Results

5.1 SYN Port Scan — SID 1000001

Nmap was used to send TCP SYN packets to all 65,535 ports on the target. The rule fires when 20 or more SYN packets originate from the same source within a 2-second window, reliably capturing both full and stealth scans.

Attack Command:

```
nmap -sS 192.168.56.101
```

Alert Generated:

```
07/01/2025-14:32:11 [**] [1:1000001:1] NMAP SYN Scan Detected [**] [Priority: 2] {TCP}  
192.168.56.102:54231 -> 192.168.56.101:80
```

Result: ✓ DETECTED

5.2 ICMP Ping Sweep — SID 1000002

Nmap's host discovery mode sends ICMP echo requests to every address in the subnet. The rule fires on 5+ echo requests from the same source within 2 seconds, effectively flagging subnet-wide ping sweeps.

Attack Command:

```
nmap -sn 192.168.56.0/24
```

Alert Generated:

```
07/01/2025-14:33:02 [**] [1:1000002:1] ICMP Ping Sweep Detected [**] [Priority: 2] {ICMP}  
192.168.56.102:0 -> 192.168.56.101:0
```

Result: ✓ DETECTED

5.3 SSH Brute Force — SID 1000003

Hydra attempted rapid credential stuffing against the SSH service using the rockyou.txt wordlist. The rule tracks connection count per source IP and fires after 5 connection attempts to port 22 within 60 seconds.

Attack Command:

```
hydra -l root -P rockyou.txt ssh://192.168.56.101
```

Alert Generated:

```
07/01/2025-14:34:45 [**] [1:1000003:1] SSH Brute Force Attempt [**] [Priority: 1] {TCP}
192.168.56.102:51001 -> 192.168.56.101:22
```

Result: ✓ DETECTED

5.4 Reverse Shell (Netcat) — SID 1000004

A simulated reverse shell payload was transmitted using Netcat. The rule performs deep packet inspection for the /bin/sh string, a hallmark of Unix-based reverse shell commands, in the raw TCP payload.

Attack Command:

```
echo "/bin/sh" | nc 192.168.56.101 4444
```

Alert Generated:

```
07/01/2025-14:36:10 [**] [1:1000004:1] Possible Reverse Shell - /bin/sh [**] [Priority:
1] {TCP} 192.168.56.102:45678 -> 192.168.56.101:4444
```

Result: ✓ DETECTED

6. Detection Summary

The table below summarises the outcome of all four attack scenarios. Every attack was successfully detected by the corresponding custom rule, with no false positives recorded during the testing period.

#	Attack Type	Tool	SID	Alert Priority	Detected
1	SYN Port Scan	Nmap	1000001	2 — Medium	YES ✓
2	ICMP Ping Sweep	Nmap	1000002	2 — Medium	YES ✓
3	SSH Brute Force	Hydra	1000003	1 — High	YES ✓
4	Reverse Shell	Netcat	1000004	1 — High	YES ✓

Overall Detection Rate	4 / 4 — 100%
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7. Defensive Recommendations

Based on the attack simulations and detection outcomes, the following hardening measures are recommended to improve the defensive posture of the monitored network:

Recommendation	Detail
Switch alert to drop rules	Convert Suricata rules from alert to drop mode to actively block confirmed threats in IPS mode, not just log them.
Deploy fail2ban alongside Suricata	Automatically ban source IPs after repeated SSH failures. Combine with Suricata alerts for layered defence.
Disable root SSH login	Set PermitRootLogin no in /etc/ssh/sshd_config. Brute-forcing a non-root account significantly reduces blast radius.
Rate-limit ICMP at the firewall	Use iptables to limit ICMP echo responses to 1/second per source, preventing effective ping sweeps.
Enable VLAN segmentation	Isolate critical services on separate VLANs so a compromised host cannot scan the full network.
Forward logs to a SIEM	Ship eve.json to Splunk, ELK, or Wazuh for long-term storage, correlation, and dashboard visualisation.

8. Conclusion

This lab successfully demonstrates the deployment and validation of a Suricata-based Network IDS in a virtualised environment. Six custom detection rules were authored from scratch, targeting the most prevalent reconnaissance and attack techniques seen in real-world network intrusions.

A 100% detection rate was achieved across all four primary scenarios with zero false positives, confirming that well-crafted signature rules remain a reliable and cost-effective layer in a defence-in-depth strategy.

Future iterations of this lab will explore anomaly-based detection using Suricata's Lua scripting engine, integration with an ELK SIEM stack, and automated response playbooks via Shuffle SOAR.

9. References

- Suricata Documentation — <https://docs.suricata.io>
- Suricata Rule Writing Guide — <https://docs.suricata.io/en/latest/rules/>
- Nmap Reference Guide — <https://nmap.org/book/man.html>
- MITRE ATT&CK; — Reconnaissance (TA0043) — <https://attack.mitre.org/tactics/TA0043/>
- MITRE ATT&CK; — Command and Scripting Interpreter: Unix Shell (T1059.004)
- CIS Benchmarks for Ubuntu Linux — <https://www.cisecurity.org/>